

ODRE PQC

Libro blanco técnico público

ODRE PQC 0.3.1 RC1 · Release 63

Alcance verificado, autoridad, límite de confianza, plataformas, FULL AUM y rendimiento medido.

Septiembre de 2026 · Base técnica Release 63

1. Autoridad y veredicto

Release 63 es la autoridad de la entrega final al cliente. El puntero activo de Production es Release 60; ambas funciones son distintas.

Authority	Verified state
Final Customer Delivery	Release 63 · g269 · 6/6 OS PASS
Active Production	Current 60 · Previous 59 · COMMITTED · Health PASS
Policy	Generation 1080 · Consumer Pin SYNCED
FULL AUM	g270 · A–R isolated PASS · destructive Production test NOT RUN BY POLICY

- Entrega técnica: Sí; seis de seis sistemas operativos oficiales PASS.
- Negative Gate 18/18 PASS; Release 63, Sealed Core y evidencia g269 permanecieron sin cambios.
- FULL AUM A–R se verificó con los bytes instalados y fixtures aislados; no se ejecutó una prueba destructiva en Production por política.

Engineering interpretation

A later PASS does not erase an earlier FAIL. Every statement is limited to the artifact, platform, environment and gate identified by its evidence.

2. Objetivo del producto

ODRE PQC vincula la ejecución protegida PQC con artefactos firmados, identidad de runtime, generación de política, consumer pin, licencia, lease y unit authority. Un requisito fallido no degrada a un fallback inseguro.

- Data Plane: Commercial Native y Sealed Core realizan las operaciones protegidas.
- Control Plane: política, pin, identidad y derecho comercial autorizan o bloquean la ejecución.
- Evidence Plane: manifest, receipt, journal y estado conservan la base de PASS, FAIL y NOT RUN.

Engineering interpretation

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3. Arquitectura de seguridad

Gateway, Native Runtime, Sealed Core, Policy, License y AUM son componentes separados unidos por una autoridad común al procesar una solicitud protegida.

- Un artefacto obligatorio ausente o modificado falla antes de instalar o aplicar.
- Un desacuerdo de policy/pin produce ejecución 0 y partial apply 0.
- Un fallo de Native/Core o protected request impide COMMIT.
- Sealed Core es inmutable, pero debe empaquetarse, instalarse, cargarse y probarse.

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4. Contrato de solicitud protegida

La solicitud autenticada se ejecuta solo después de validar derecho, identidad, sincronización de policy, Native readiness y carga del Core.

- El handler se ejecuta exactamente una vez por cada solicitud protegida exitosa.
- Unauthorized execution, plaintext fallback y downgrade clásico implícito son cero.
- Un código health aislado no basta; protected request forma parte del gate de commit.

Engineering interpretation

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5. Cadena de confianza de Release

La selección usa la autoridad firmada y SHA-256 exacto, nunca el nombre mayor ni la fecha más reciente.

Generation	Role
60	Active Production baseline
61	Ubuntu runtime-manifest pin repair lineage
62	Immutable intermediate successor
63	Final Customer Delivery authority

- Archivo → inventory/manifest → envelope/CMS → installer → carga instalada → integración protegida → sello ZIP.
- Release 60/61/62 no se sobrescribieron; Release 63 es un sucesor monotónico.
- Installer execution, installation integration, runtime integration y Customer Delivery son gates distintos.

Engineering interpretation

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6. Ciclo FULL AUM

DISCOVER, VERIFY, STAGE, SWITCH, HEALTH, COMMIT, ROLLBACK y RECOVERY forman una transacción durable.

- Single writer lock evita dos apply simultáneos.
- Un candidate parcialmente copiado no pasa a VERIFIED.
- Estado atómico y journal durable evitan JSON parcial o COMMITTED falso.
- Se comprueba capacidad antes de staging y no se borran activos de rollback.

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7. Servicios y administración

Los servicios Windows usan el lifecycle de SCM y `start_server` no ejecuta directamente `exe/python` para eludirlo.

- No se reinicia un servicio ya activo; solo se inicia uno detenido mediante SCM.
- Admin Update es autenticado, dinámico y read-only.
- Production: current 60, previous 59, floor 60, policy 1080, pin SYNCED, COMMITTED/PASS, health 8000/18443 HTTP 200.

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8. Cobertura oficial

El mismo contrato pasó en Windows Server 2022, Ubuntu 22.04/24.04, Debian 12, Rocky Linux 9.8 y AlmaLinux 9.8.

OS	Result	Receipt SHA-256
Windows Server 2022	PASS	5E390E2062034FDE03BFA304D6A64EEAA191F79CB292E0EA913086CF5CB30489
Ubuntu 22.04	PASS	2E19C70F2992A1FA520E757DFC64A2FDB78DDEDBAB4E6AA4F943954D7B6E627D
Ubuntu 24.04	PASS	4AFBB78E6AB65D7D9389FA2B337C7E7350828AD69D827011CA399BAB226BC1B4
Debian 12	PASS	2597605241A07F53282EE0CEE6E5EF101919E178AAD00CBC06FF7FEBB2060293
Rocky Linux 9.8	PASS	701244BC819DA109AA190621F83EC91C36C5EDC0129041D80509F9AA952070AE
AlmaLinux 9.8	PASS	EBDBA21B870BD33032C26420669FC3F313890F45548B3369EB974C9E8FEB8563

- Los adaptadores SCM/ACL o service/permission no cambian el trust model.
- Containers, Kubernetes y otras versiones de Windows/Linux no forman parte de este veredicto.

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9. Fail-closed y pruebas negativas

ZIP/SHA alterado, fallo CMS, inventory mismatch, policy drift, generaciones duplicadas o inferiores, fallo runtime y falta de espacio deben detenerse con seguridad.

- REJECTED_PRE_APPLY implica Production change 0.
- Un fallo health tras switch restaura Product, policy, pins y runtime anteriores.
- La evidencia FAIL histórica se conserva tras la corrección y el retest.

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10. Evidencia e interpretación

g269 demuestra Final Customer Delivery; g270 demuestra FULL AUM acotado y Admin live read-only.

- Updater 208/208, drop-in 14/14 y 12-A/Admin 19/19 PASS.
- Drive upload/readback y dos Notion append/readback PASS; re-download SHA NOT RUN por política.
- No se declara certificación FIPS 140-3/Common Criteria ni rendimiento universal.

Engineering interpretation

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Rendimiento medido

Medición histórica g108 en Windows Server 2022, loopback y candidate. No se repitió en Final Release 63 y no se generaliza a WAN ni hardware del cliente.

Test	Measured value
Environment	Windows Server 2022 build 20348 · AMD EPYC 9454P · 4 cores · 16 GiB · loopback
Protected 1W max stable mean	1,176.9337 RPS (range 1,163.8582–1,190.0091)
Baseline 1W max stable mean	3,442.9391 RPS (range 3,413.4320–3,472.4462)
Same concurrency 2	Baseline 3,442.9391 / Protected 1,104.7263 RPS · reduction 67.9133%
Latency c2 baseline → protected	p50 0.5468→1.7223 ms · p95 0.7238→2.2881 ms · p99 0.9382→2.6597 ms
Protected security	200,128 success / 200,128 handler · invariant failure 0 · plaintext HTTP 426 · fallback NONE
Protected 2W / 4W	NOT_MEASURED · single_worker_lease_unavailable fail-closed
Baseline scaling reference	2W 5,035.1443 RPS c8 · 4W 5,475.0515 RPS c32 · zero error

- Same business_handler, POST /secure/echo and serialization were used for Baseline and Protected.
- The 2W/4W protected values were not estimated or multiplied.
- These figures do not support a claim that PQC overhead is negligible.

Conclusion

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Release 63 Final Customer Delivery is technically PASS. Active Production Release 60 remains a separate committed authority. Unexecuted scopes remain explicitly NOT RUN.